DC Operating-Point Diagnosis & Fix — Two-Stage CMOS LDO Error Amplifier

0.18 µm CMOS (gpdk180) • PMOS-input two-stage Miller op-amp • analysis of the attached Cadence DC op-point screenshot. All operating-point numbers below are read directly from the screenshot; assumptions and recommendations are labelled.

> **Headline result.** The PMOS current-mirror gate net (shared by M5, M7, M8) is at **0 V** — every mirror device reads **V_{GS} = -1.8 V** (gate = ground). This forces $|V_{SG}|$=1.8 V, an enormous overdrive $|V_{OV}|$≈1.28 V that the few-mV to 174-mV available $|V_{SD}|$ cannot satisfy → M5/M7/M8 collapse into triode (region 1), and the supply draws **110.4 µA** (~199 µW). Root cause is a **wiring error: M8 is not diode-connected** (its gate/mirror line is tied to ground instead of to its own drain). Secondary issue: input common-mode 0.9 V is too high for tail headroom.

1. Screenshot Interpretation (Topology)

Two-stage Miller-compensated op-amp with **PMOS differential input pair**:

- **M1, M2** — PMOS differential-pair input devices (gates = Vin- / Vin+). Sources tied to tail node (1.76871 V).
- **M3, M4** — NMOS current-mirror active load (M3 diode-connected, M4 mirror output). Sources = gnd.
- **M5** — PMOS tail current source (source = VDD, drain = tail node).
- **M8** — PMOS bias/reference device fed by Iref (intended diode-connected mirror master).
- **M6** — NMOS second-stage common-source gain transistor (drain = Vout).
- **M7** — PMOS second-stage current-source load (source = VDD, drain = Vout).
- **Cc + Rz** — Miller compensation cap with nulling resistor, from first-stage output node to Vout (Rz shows v=0,i=0 → correctly carries no DC).
- **CL** — load capacitor at Vout. **Iref** — bias current source into M8. **Vdd** = 1.8 V.

Bias intent: M8 (1 µA) is the mirror master; M5 and M7 mirror it (observed width ratios ≈ 1 : 10 : 100 from the triode currents). Gate net of M5/M7/M8 is common — but it is grounded, not driven by M8's drain.

2. Extracted Transistor Parameters (read from screenshot)

MOS	Type	ID	VGS	VDS	gm	Region
M1	PMOS	-4.89551 µA	-868.713 mV	-857.086 mV	21.3877 µS	2 (sat)
M2	PMOS	-4.89551 µA	-868.713 mV	-857.086 mV	21.3877 µS	2 (sat)
M3	NMOS	4.89552 µA	911.627 mV	911.627 mV	18.2728 µS	2 (sat)
M4	NMOS	4.89552 µA	911.627 mV	911.627 mV	18.2728 µS	2 (sat)
M5	PMOS	-9.79102 µA	-1.8 V	-31.287 mV	4.45577 µS	**1 (triode)**
M6	NMOS	99.576 µA	911.627 mV	1.62606 V	296.911 µS	2 (sat)
M7	PMOS	-99.5759 µA	-1.8 V	-173.936 mV	50.8145 µS	**1 (triode)**
M8	PMOS	-1 µA	-1.8 V	-3.15142 mV	445.181 nS	**1 (triode)**

Sources: Vdd i = -110.367 µA (supply delivers 110.367 µA) • Iref v = 1.79685 V. **Current balance:** tail 9.79 + 2nd-stage 99.58 + bias 1.0 = 110.37 µA. **Power = 1.8 V × 110.367 µA = 198.7 µW.**

3. Extracted W/L Values

> **W/L not reliably legible.** The red annotations are ID/VGS/VDS/gm/region and the cyan numbers are DC node voltages; the device W and L fields are not clearly readable in this screenshot (visible hints: "900m" near M1 suggests W≈0.9 µm; other "1.8"/"0" tokens are ambiguous). Please confirm the exact W/L from the schematic property editor. Below is what can be *inferred*, not measured.

MOS	Type	W (to confirm)	L (to confirm)	W/L	Inference basis
M1,M2	P	~0.9 µm	?	?	"900m" label

M3,M4	N	0.400 µm	0.2855 µm	1.401	stated modification
M8 : M5 : M7	P	width ratio ≈ 1 : 10 : 100			from triode ID ratio at equal |VGS|=1.8 V

4. M3/M4 W/L Modification Verification

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Original: 280 nm / 200 nm = 1.400   |   Modified: 400 nm / 285.5 nm = 1.4011 ≈ 1.4 — verified.
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> **Simulation confirms the change is behaving correctly:** M3 and M4 both carry **4.89552 µA** (equal to the M1/M2 branch current 4.89551 µA) and are in saturation (region 2). The 1:1 mirror is accurate.

But equal W/L does NOT make the device electrically identical. Going 200 nm → 285.5 nm (L up 43 %, W up 43 %):

Quantity	Effect of longer L (same W/L)
Drain current (same VGS)	Approx. preserved to first order (same W/L, same V_{OV}), but shifts slightly because λ, DIBL and velocity saturation differ at longer L
Channel-length modulation λ	Decreases (λ ∝ 1/L) — less current variation with V_{DS}
Output resistance $r_o=1/(\lambda I_D)$	Increases → higher first-stage gain
g_m	~unchanged at fixed I_D/W/L to first order; slightly lower if velocity-saturation relief changes I_D
Parasitic caps C_{gs},C_{gd},C_{db}	Increase (~WL area up 2×) → lower bandwidth, more mirror settling time
Matching ($\sigma_{VT} \propto 1/\sqrt{WL}$)	Improves (larger area) → lower mirror offset
Current-mirror accuracy	Improves (higher r_o, better matching)

Net: the modification is beneficial for gain, matching and mirror accuracy, at a small cost in bandwidth/area. Behaving as expected.

5. Current Operating-Region Analysis (all devices)

gpdk180 uses BSIM3v3; the Spectre op-point **region** code convention is **0=cutoff/subthreshold, 1=triode(linear), 2=saturation, 3=subthreshold, 4=breakdown**. So region 2 = saturation, region 1 = triode. (Confirm via ADE → Results → Print → DC Operating Points legend for your model version — numbering is not universal.)

MOS	Condition checked	Calculated region	Cadence region	Match	Comment
M1/M2 (P)	$\|V_{SG}\|=0.869>\|V_{th}\|$; $\|V_{SD}\|=0.857\geq\|V_{OV}\|$	saturation	2	Yes	Healthy input pair
M3/M4 (N)	$V_{GS}=0.912>V_{th}$; diode $V_{DS}=V_{GS}$	saturation	2	Yes	Mirror OK, I=4.9 µA
M6 (N)	$V_{GS}=0.912$; $V_{DS}=1.626$ large	saturation	2	Yes	Sat but huge I=99.6 µA
M5 (P)	$\|V_{SD}\|=0.031 \ll \|V_{OV}\|=1.28$	triode	1	Yes	Gate grounded
M7 (P)	$\|V_{SD}\|=0.174 \ll \|V_{OV}\|=1.28$	triode	1	Yes	Gate grounded
M8 (P)	$\|V_{SD}\|=0.003 \ll \|V_{OV}\|=1.28$	triode	1	Yes	Not diode-connected

6. Why M5 / M7 / M8 Are in the Linear Region — Mathematical Proof

PMOS saturation requires $|V_{SD}| \geq |V_{OV}|$ where $|V_{OV}| = |V_{SG}| - |V_{TH}|$. Using **assumed** $|V_{TH,p}| = 0.518$ V (from the paper's Table 1; **gpdk180 actual must be extracted**, typically 0.42-0.50 V):

MOS	|VSG|	|VTH|(assumed)	|VOV|	Actual |VSD|	Required |VSD|	Saturation?
M5	1.800 V	0.518 V	1.282 V	0.0313 V	≥ 1.282 V	**NO** (41× short)
M7	1.800 V	0.518 V	1.282 V	0.1739 V	≥ 1.282 V	**NO** (7.4× short)
M8	1.800 V	0.518 V	1.282 V	0.0032 V	≥ 1.282 V	**NO** (407× short)

Even with the smallest realistic $|V_{TH}|$=0.42 V, $|V_{OV}|$=1.38 V — still far above the available $|V_{SD}|$. The devices are physically incapable of saturating with the gate at ground. The $|V_{SG}|$=1.8 V (not the drain voltage) is the culprit.

7. DC Bias Path Trace

Node	M5 (tail)	M7 (2nd-stage load)	M8 (bias master)
Source	1.8 V (VDD)	1.8 V (VDD)	1.8 V (VDD)
Gate	**0 V**	**0 V**	**0 V**
Drain	1.76871 V (tail node)	1.62606 V (Vout)	1.79685 V (Iref node)
|VSD|	31.3 mV	173.9 mV	3.2 mV

The three PMOS gates are all at 0 V. If M8 were correctly diode-connected, its gate would equal its drain (1.79685 V) giving $|V_{SG,8}|$≈3 mV — but the readout shows -1.8 V, proving the shared gate line is tied to **ground**, not to M8's drain. Because the drains of M5/M7 are pinned high by the first/second-stage nodes (tail source = $V_{CM}+|V_{SG1}|$ = 0.9+0.869 = 1.769 V), there is almost no room below VDD, so with any large $|V_{SG}|$ the devices sit in triode.

8. Root Cause

PRIMARY (necessary fix): M8 is not diode-connected — the PMOS mirror gate net is grounded. This single fault causes (a) $|V_{SG}|$=1.8 V on M5/M7/M8, (b) triode operation of all three, and (c) the runaway ~110 µA supply current.
SECONDARY (also necessary): input common-mode V_{CM}=0.9 V is too high. It pushes the tail node to 1.769 V, leaving only 31 mV for M5 — so even after the mirror is fixed, M5 would still lack headroom. V_{CM} must drop to ≈0.5-0.55 V (which also matches V_{REF}).
TERTIARY (optimization): mirror width ratios (~1:10:100) make the 2nd stage draw ~100 µA → ~199 µW, far above the µW-class harvesting budget. Reduce M5/M7 widths after the DC point is valid.

It is NOT primarily a W/L problem, NOT a supply problem, and NOT an Iref problem. It is a mis-wire plus a common-mode choice.

9. Required Circuit Changes (prioritized)

A. Necessary

#	Change	Device	Why / equation	Direction & start value	Effect
A1	Diode-connect the mirror	M8 (+ shared gate net M5/M7)	Mirror needs gate = master drain so $\|V_{SG}\|$ self-sets for I_{ref}: $I=½\mu C_{ox}(W/L)V_{OV}^2$ → $\|V_{SG}\|$≈0.6-0.7 V for 1 µA	Wire M8 gate to M8 drain (Iref node); keep M5,M7 gates on that net	$\|V_{SG}\|$↓ to ~0.65 V, $\|V_{OV}\|$↓ to ~0.15 V, currents collapse to mirror ratio×I_{ref}, M5/M7/M8 → saturation
A2	Lower input common-mode	Vin+ , Vin- DC	Tail headroom: $\|V_{SD5}\|$ = VDD - ($V_{CM}+\|V_{SG1}\|$) ≥ $\|V_{OV5}\|$. Need V_{CM} ≤ VDD - $\|V_{SG1}\|$ - $\|V_{OV5}\|$	Reduce 0.9 V → 0.50-0.55 V	Tail node ↓ to ~1.15 V, $\|V_{SD5}\|$ ≈0.63 V → M5 solidly saturated; also aligns with V_{REF}

B. Optional optimization (after A, for power/spec)

#	Change	Device	Why	Effect
B1	Reduce mirror ratio / width	M5, M7	Cut 2nd-stage 100 µA to a few µA for the µW harvesting budget	I↓, power↓; g_{m6}↓ and UGB↓ (retune Cc); r_o↑ gain↑
B2	Increase M1/M2 W (lower V_{OV1})	M1, M2	$\|V_{SG1}\|$=0.869 V is high (V_{OV1}≈0.35 V); lowering it adds tail headroom and gm/I efficiency	$\|V_{SG1}\|$↓, more headroom, g_{m1}↑ (UGB↑), caps↑
B3	Re-tune Cc, Rz	Cc, Rz	After currents change, set UGB=$g_{m1}/2\pi$Cc and Rz=$1/g_{m6}$	Sets phase margin/stability

Do A1 first, re-run DC OP, then A2, re-run, then optimize. Each change is isolated so you can see which fixes what.

10. Revised W/L / Bias Recommendations

Recommendations (verify in Cadence), low-power strong-inversion targets. Keep I_{ref}=1 µA. Mirror ratios chosen for a modest, harvesting-friendly current.

MOS	Type	Rec. W	Rec. L	W/L	Exp. ID	Exp. |VOV|	Exp. region
M1,M2	P	2.0 µm	1 µm	2.0	1 µA	0.12 V	sat
M3,M4	N	0.40 µm	0.2855 µm	1.40	1 µA	0.15 V	sat
M5 (tail)	P	1.6 µm	1 µm	1.6	2 µA	0.20 V	sat
M6	N	1.2 µm	1 µm	1.2	4 µA	0.15 V	sat
M7	P	3.2 µm	1 µm	3.2	4 µA	0.20 V	sat
M8	P	0.8 µm	1 µm	0.8	1 µA	0.20 V	sat (diode)

Total I ≈ 2 (tail) + 4 (2nd stage) + 1 (bias) = 7 µA → P ≈ 12.6 µW (vs 199 µW now). Ratios: M5:M8=2, M7:M8=4. Systematic-offset match: $(W/L)_6/(W/L)_4 = 2(W/L)_7/(W/L)_5$ → 1.2/1.4=0.857 vs 2×3.2/1.6=4.0 — not matched, so expect some systematic offset; adjust M6 or M7 to balance if needed (this is a second-order optimization).

11. DC Operating-Point Analysis Procedure

1. Open the schematic; Check & Save (fix the M8 diode wire first).
2. Launch ADE-L (Analog Design Environment).
3. Set Vdd = 1.8 V (vdc).
4. Set Vin+ DC = **0.5 V**, Vin- DC = **0.5 V** (common-mode = V_{REF}; do NOT use 0.9 V — it starves the tail).
5. Set Iref = 1 µA (idc), correct polarity (sinks from M8 drain to gnd).
6. Analyses → Choose → **dc**, enable "Save DC Operating Point". Leave sweep off.
7. Outputs → Save All (or select device currents).
8. Run.
9. Results → Annotate → DC Operating Points (values appear on schematic).
10. For each MOS read ID, VGS, VDS, gm, region; confirm all region = 2.
11. Check every M1-M8 before any AC.

Why V_{CM}=0.5 V: PMOS pair needs $V_{CM} \le VDD - |V_{SG1}| - |V_{OV,tail}| \approx 1.8 - 0.65 - 0.2 = 0.95$ V max, and can go down to ~0.1 V. 0.5 V sits comfortably mid-range and equals V_{REF} for the LDO.

12. DC Common-Mode Sweep Procedure

1. Tie Vin+ and Vin- to a single source V_{CM} (or two sources with identical value driven by a variable).
2. Analyses → dc → sweep **V_{CM} from 0 to 1.8 V**, step 10 mV.
3. Outputs: plot region/VDS of M1, M2, M5 and the tail-node voltage, and the branch currents.
4. Read: **ICMR** = the V_{CM} band where M1/M2 stay in saturation and M5 keeps $|V_{SD}| \ge |V_{OV}|$.
5. Upper limit: where M5 (tail) enters triode (V_{CM} too high). Lower limit: where M1/M2 leave saturation (V_{CM} too low, drains pinch).
6. Confirm bias currents stay ~constant across the usable band; watch output node move.

Expected usable ICMR ≈ 0.1 V to ~0.95 V; pick operating V_{CM}=0.5 V near the center.

13. AC Analysis Procedure

Only after a valid DC point (all region 2). Configure a **differential** AC drive so the DC operating point is untouched and only a small signal is injected:

Input	DC	AC magnitude	AC phase
Vin+ (M2 gate)	0.5 V	0.5 (or 1)	0°
Vin- (M1 gate)	0.5 V	0.5 (or 1)	180°

This puts $+V_{ac}/2$ and $-V_{ac}/2$ on the two gates (total differential = 1), keeping common-mode fixed. **Alternative:** AC=1 on one input, AC=0 on the other — simpler, and since the pair is differential the small-signal gain to Vout is essentially the same; use this for a quick single-ended read. Do not confuse this AC magnitude (small-signal perturbation) with the 0.5 V DC (which sets the operating point).

- Analyses → ac → sweep 1 Hz to ≥ 10× expected UGB (e.g. 1 Hz – 100 MHz), logarithmic, points/decade ≥ 20.

- **DC gain** = dB20(V("/Vout")) at low frequency.
- **UGB** = cross(dB20(V("/Vout")),0,1) — where gain = 0 dB.
- **Phase margin** = 180° + phase(V("/Vout")) at UGB (use phaseMargin() function).
- **Dominant pole** = -3 dB corner; **non-dominant pole** near the gain roll-off / phase second break.

14. Transient / Slew-Rate Analysis

Configure the amp in unity-gain (or the intended feedback) and apply a step:

- Input: **vpulse** on Vin+, DC bias 0.5 V, pulse to 0.6 V (100 mV small step for settling) and separately a larger step (e.g. 0.3 V) for slew.
- Rise/fall time = 1 ns (fast edge), pulse width = 20 µs, period = 40 µs.
- Load: CL as in schematic (the pass-gate cap, ~1-2 pF for standalone char).
- Run tran to ~50 µs.
- **SR+** = max positive dVout/dt on the rising edge; **SR-** = max negative dVout/dt on the falling edge (use the calculator slewRate() or deriv()).
- Settling time = time to within 1 % of final; check for ringing (relates to phase margin).

15. Power Analysis

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P = VDD × I_supply. Measured I_supply = 110.367 µA → P = 1.8 × 110.367 µ = 198.7 µW.
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Read I_supply directly from the Vdd source current (screenshot: i = -110.367 µA). **This is ~20-30× too high** for an energy-harvesting PMU error amp and is dominated by the 99.6 µA second stage caused by the mirror fault. After fixes A+B and optimization B1, expect ~7-15 µA → ~13-27 µW. Do NOT target the paper's 36 nW (that is a subthreshold PMU-total figure); a conventional-inversion 0.18 µm error amp in the low-tens-of-µW is appropriate.

16. Corner & Temperature Analysis

After the nominal design passes: run ADE-XL with gpdk180 corner models **TT, SS, FF, SF, FS** × temperatures **-40, 27, 85 °C**. At each corner check: DC gain, UGB, phase margin, slew rate, power, ICMR, output swing, and that every device stays in saturation. Worst cases: SS/85°C (lowest gain/UGB), FF/-40°C (highest power, lowest PM), SF/FS (input-pair & mirror mismatch → offset). Pass: gain ≥ 55 dB, PM ≥ 45°, all devices region 2.

17. Hand-Calculation vs Cadence Comparison

Extracting effective $|V_{OV}|$ from $gm = 2I_D/V_{OV}$ and comparing with V_{GS}-V_{TH} (assumed $|V_{TH,p}|$=0.518, $V_{TH,n}$=0.462):

MOS	|VOV| from 2ID/gm	|VOV| from VGS-VTH	Error %	Reason for gap
M1 (P)	2×4.8955/21.388=0.458 V	0.869-0.518=0.351 V	30 %	gm below square law (mobility deg., short channel)
M3 (N)	2×4.8955/18.273=0.536 V	0.912-0.462=0.450 V	19 %	same; body effect on V_{TH}
M6 (N)	2×99.576/296.91=0.671 V	0.912-0.462=0.450 V	49 %	strong velocity saturation at high I

```
Error(%) = |sim − calc| / |calc| × 100
```

Hand equations ($ID=\frac{1}{2}\mu C_{ox}(W/L)V_{OV}^2$, $gm=2ID/V_{OV}$) are first-order only. Real BSIM3 results differ due to: channel-length modulation, velocity saturation, mobility degradation, body effect, DIBL, finite r_o, and parasitics. **gpdk180 V_{TH}, μC_{ox}, λ must be extracted from the DC op-point** (read V_{TH} and gm directly rather than assuming) for accurate hand checks. Do not use subthreshold equations — this design is conventional inversion.

18. LDO Error-Amplifier Requirements

Parameter	Effect on the LDO
DC gain	Sets line & load regulation (∝ 1/loop gain); want ≥ 60 dB
UGB	Loop transient-recovery speed; ~1 MHz adequate with large output cap
Phase margin	Loop stability; keep amp PM high (~60-70°) so total loop PM ≥ 45°

Slew rate	How fast the pass-gate can be driven on load steps
Output resistance	Sets 2nd-stage gain; interacts with pass-gate load
ICMR	Must include V_{REF}=0.5 V comfortably (PMOS pair does)
Output swing	**Critical:** must drive the PMOS pass gate from near VDD (pass off) down to VDD-$\|V_{SG,pass}\|$ (pass fully on)
Power	Directly reduces PMU efficiency → keep in low µW

The op-amp output (Vout node, currently 1.626 V) drives the PMOS pass-transistor gate. For an LDO delivering up to ~1 mA, the amp must swing its output over roughly **1.0-1.75 V** (with VDD=1.8) to modulate the pass $|V_{SG}|$ from cutoff to full conduction. This is exactly why M6 (pull-down) and M7 (pull-up) must both be in saturation with adequate swing — the present M7 triode state cannot pull the gate high cleanly. Fixing M7 is therefore also an LDO-function requirement, not just a bias nicety.

19. Final Recommended Design Summary

Parameter	Target	Current simulation	Recommended
VDD	1.8 V	1.8 V	1.8 V
VCM (inputs)	≈ V_{REF}	0.9 V (too high)	0.50 V
Iref	1 µA	1 µA	1 µA
Total current	low µA	110.4 µA	~7 µA
Power	≤ ~15 µW	198.7 µW	~12.6 µW
DC gain	≥ 60 dB	(invalid DC pt)	≥ 60 dB
UGB	~1 MHz	-	~1 MHz
Phase margin	≥ 60°	-	~65°
Slew rate	≥ 0.5 V/µs	-	~1 V/µs
CL	1-2 pF	as drawn	1-2 pF
Cc	~1 pF	as drawn	retune to $g_{m1}/2\pi UGB$

(Recommended per-device W/L is in Section 10.)

20. Final Cadence Verification Checklist / Workflow

```
SCHEMATIC  -->  FIX M8 diode wire (gate to drain, shared to M5,M7 gates)
   |            Cadence: add wire; Check&Save. Pass: M8 VGS != -1.8
CHECK W/L  -->  Confirm W,L in property editor vs Section 10
   |
DC OP PT   -->  VCM=0.5V, Iref=1uA. Analyses>dc>Save DC Op Pt. Run.
   |            Pass: ALL M1-M8 region=2; Isupply ~ single-digit uA
CHECK REGIONS --> Annotate>DC Operating Points. Pass: no region=1
   |
FIX M5/M7/M8 --> if still triode: lower VCM further / reduce M5,M7 W
   |            Re-run DC. Pass: |VSD| >= |VOV| each PMOS
CM SWEEP   -->  dc sweep VCM 0..1.8V. Observe ICMR. Pass: 0.5V mid-band
   |
AC         -->  diff AC (0/180 deg), 1Hz..100MHz log.
   |            Pass: gain>=60dB, UGB~1MHz, PM>=60deg
TRANSIENT  -->  vpulse step, unity gain. Pass: SR ok, settle <1%, low ring
   |
POWER      -->  P=VDD*I(Vdd). Pass: ~10-20uW
   |
CORNERS    -->  ADE-XL TT/SS/FF/SF/FS. Pass: gain>=55dB, PM>=45deg
   |
TEMPERATURE --> -40/27/85C. Pass: regions hold, specs in range
   |
FINAL W/L OPT --> trim for offset/gain/PM balance. Freeze.
```

> **Immediate next action:** (1) wire M8 as a diode (gate→drain), (2) set V_{CM}=0.5 V, (3) re-run DC op-point. Those two changes alone should move M5/M7/M8 into saturation and cut the supply current by ~10-20×. Everything else is

optimization on top of a valid DC point.